

SAMIR SEN

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PROFESSIONAL SUMMARY

Highly motivated and detail-oriented Artificial Intelligence and Software Engineering enthusiast. With a strong academic background and hands-on experience in machine learning, backend development, and cloud computing, I am dedicated to building scalable AI-driven solutions. Seeking a challenging role to leverage my technical skills in Python, Java, and cloud architecture to solve complex real-world problems.

TECHNICAL PROFICIENCY

- **Machine Learning:** Supervised & Unsupervised Learning, Regression, Classification (Random Forest), Deep Learning (CNN, SSD MobileNetV2), Data Preprocessing, Feature Engineering, Model Evaluation.
- **Backend & API:** Python, Java, Spring Boot, FastAPI, RESTful API design, Microservices architecture, Backend-AI model integration.
- **Data & Tools:** Pandas, scikit-learn, TensorFlow, OpenCV, Jupyter Notebook, MySQL, Oracle, Git, GitHub.
- **Cloud/DevOps:** AWS Cloud Practitioner, Azure fundamentals, Docker containerization, Linux (Bash scripting), CI/CD basics.

KEY PROJECTS

Real-Time Computer Vision Pipeline

Tech Stack: SSD MobileNetV2, OpenCV, FastAPI, Spring Boot, AWS

- Engineered an end-to-end real-time object detection system capable of processing live video feeds for classification.
- Implemented SSD MobileNetV2 for lightweight yet accurate detection, achieving consistent frame rates.
- Designed a robust, scalable backend using Spring Boot for user management and FastAPI for high-throughput model inference.
- Deployed the solution in a containerized environment using Docker to ensure portability and ease of maintenance.

Predictive Meteorological Modeling

Tech Stack: Python, Pandas, Scikit-Learn, Random Forest

Developed multi-variable weather forecasting models by analyzing complex meteorological datasets.

- Performed extensive feature engineering, including time-series normalization and anomaly detection, to improve model robustness.
- Utilized Random Forest and Linear Regression algorithms to predict weather metrics, evaluating performance through RMSE and MAE metrics.
- Documented the research pipeline and model results in a comprehensive Jupyter Notebook, showcasing data visualization techniques.

Salary Insights Data Analytics

Tech Stack: Python, Pandas, Matplotlib, Seaborn

- Conducted exploratory data analysis (EDA) on salary datasets to identify hidden trends and correlations between job roles, experience, and compensation.
- Cleaned and pre-processed unstructured raw data, handling missing values and outliers to ensure high-quality analysis.
- Created intuitive visualizations to highlight key factors influencing salary distribution, enabling data-driven insights.
- Presented findings clearly to provide recommendations for professionals and hiring stakeholders.

EDUCATION

BSc (Hons) Computing (Artificial Intelligence)

Islington College (London Metropolitan University) | 2022-2025

- **Relevant Coursework:** AI & Machine Learning, Advanced Algorithms, Data Structures, Cloud Computing, System Architecture.
- Engaged in collaborative, agile-based group projects throughout the degree.

CERTIFICATIONS

- **AWS Cloud Quest: Cloud Practitioner (July 2023):** Demonstrated proficiency in cloud services, security, and architectural best practices.

SOFT SKILLS & PROFESSIONALISM

- **Analytical Problem Solving:** Proven track record of breaking down complex system requirements into modular, manageable components.

- **Collaborative Leadership:** Experienced in team-based environments, coordinating effectively with peers on technical deliverables.
- **Adaptability & Learning:** Quick to adopt new technologies and programming paradigms, evidenced by a broad tech stack spanning Java to Python-based AI.
- **Communication:** Able to articulate technical concepts to non-technical stakeholders clearly and concisely.